

ONLINE VOTING SYSTEM WITH AADHAAR-STYLE OTP VERIFICATION

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Team No.: 9 | Section: 12 | KLH Bachupally Campus, CSE | PBL 2026-27

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1. ABSTRACT

Online Voting System is a web-based application designed to modernize the traditional voting process. It provides a convenient platform for voters to cast their votes digitally, while OTP-based verification adds an additional layer of voter authentication. The system uses a centralized database to manage voters, candidates, elections and votes, while administrations manage elections and generate results efficiently.



KEY IDEA

Secure Authentication + Database Management + Online Voting

2. PROBLEM STATEMENT

TRADITIONAL VOTING REQUIRES

- Physical presence at polling locations
- Manual voter verification
- Manual record maintenance
- Manual vote counting
- Greater election-management effort

ONLINE VOTING CHALLENGES

- Authorized voter access
- Duplicate voting prevention
- Accurate voting records
- Vote integrity
- Secure voter and election data

Therefore, there is a need for a secure and user-friendly online voting system that provides OTP-based voter authentication, maintains vote integrity, securely manages voter and election data, and enables efficient election and result management.

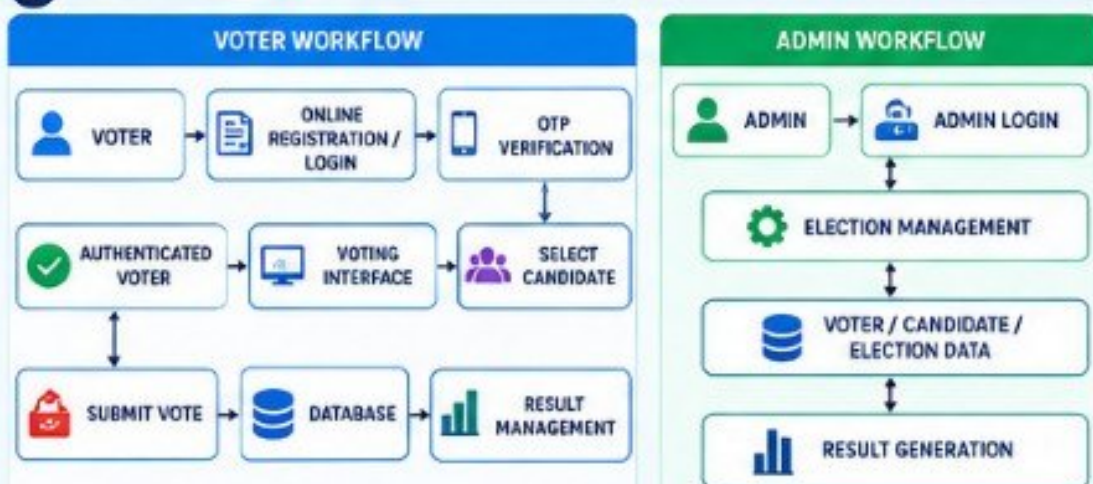
3. OBJECTIVES

- 01 Authenticate voters using OTP verification.
- 02 Provide a simple online voting interface.
- 03 Allow only verified voters to cast their vote.
- 04 Prevent multiple voting in the same election.
- 05 Maintain accurate voter and voting records.
- 06 Simplify election and result management.

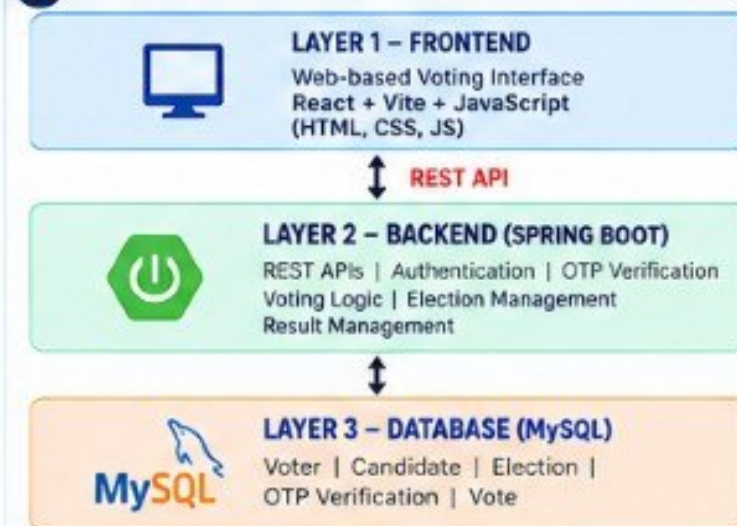
4. EXISTING vs PROPOSED SYSTEM

Feature	Existing / Traditional System	Proposed Online Voting System
Voting mode	Physical polling	Digital voting interface
Voter verification	Manual verification	OTP-based authentication
Record maintenance	Manual record maintenance	Centralized database
Vote counting	Manual vote counting	Automated vote / result management
Election management	More effort in election management	Controlled voting access and duplicate voting prevention

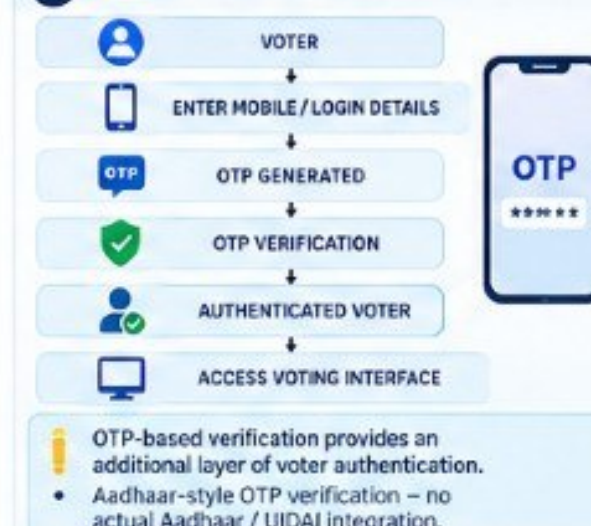
5. PROPOSED SYSTEM



6. SYSTEM ARCHITECTURE



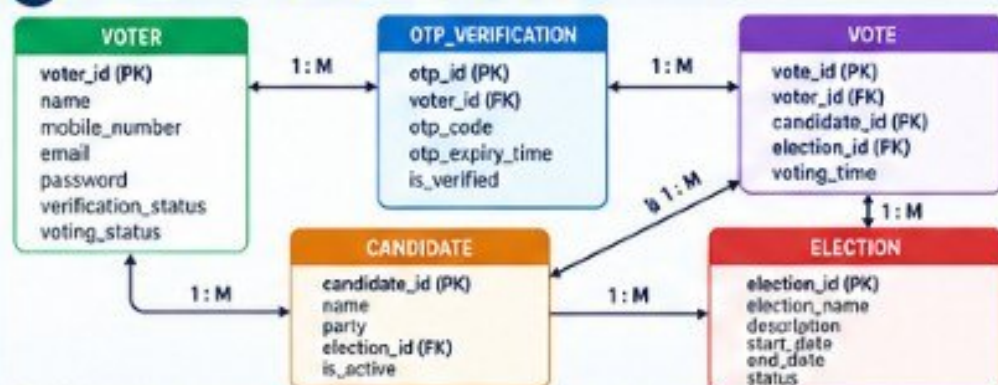
7. OTP VERIFICATION



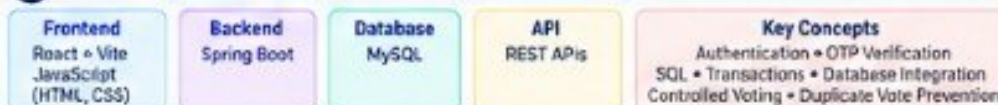
8. VOTING WORKFLOW

- 01 Voter accesses system
- 02 OTP authentication
- 03 Voter verification
- 04 Voting interface displayed
- 05 Candidate selected
- 06 Vote submitted
- 07 Voting record stored in MySQL
- 08 Duplicate voting prevented
- 09 Results managed by administrator

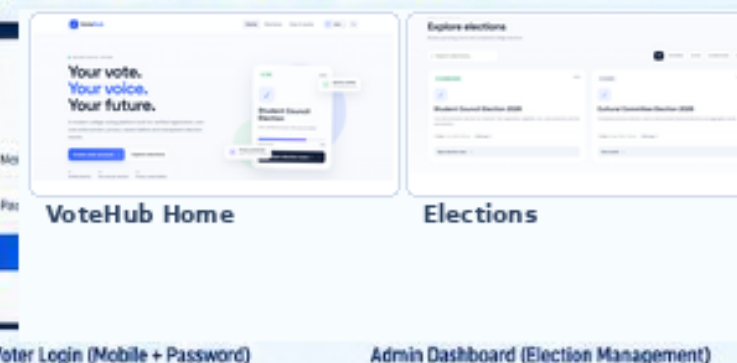
9. DATABASE DESIGN – ER OVERVIEW



13. TECHNOLOGY STACK



10. IMPLEMENTATION SCREENSHOTS



11. RESULTS & TESTING

TEST AREAS	EXPECTED OUTCOMES
✓ OTP authentication	Only verified voters can access voting.
✓ Verified voter access	Multiple voting is prevented.
✓ Voting interface	Voting records are maintained accurately.
✓ Duplicate voting prevention	Election and result management is simplified.
✓ Voter records	MySQL stores voting-related information centrally.
✓ Candidate records	
✓ Election records	
✓ Vote records	
✓ Database integration	
✓ Result management	

12. SECURITY & DATA INTEGRITY

- ✓ OTP-Based Authentication
- ✓ Secure Voting Access
- ✓ Verified Voters Only
- ✓ Duplicate Voting Prevention
- ✓ Centralized Database
- ✓ Accurate Voting Records
- ✓ Controlled Election Management

14. CONCLUSION

The Online Voting System provides a structured digital voting workflow using OTP-based voter authentication, Spring Boot backend, MySQL relational database, controlled voting access, duplicate voting prevention and automated vote and result management. The project demonstrates practical concepts from database engineering and backend development, SQL, transactions, REST APIs, authentication and database integration.

15. FUTURE SCOPE

- Enhanced voter verification
- Real-time election monitoring
- Improved result analytics
- Scalable deployment for larger elections
- Integration with secure notification services

16. REFERENCES

1. Spring Boot Documentation
2. MySQL Documentation
3. REST API Documentation
4. React / Web Development Documentation

17. GITHUB

DBS-DS_PROJECT
PROJECT_OnlineVotingSystem

https://github.com/Jhanasri-thota/DBS-DS_PROJECT

